Verification Goal & Edge Cases	Expected Results (Registers & FSM)	Input Sequence & Timing	Instruction / Action	Category	ID	Module
Safety: Verifies hardware drops to a known zero-state immediately, preventing rogue execution.	Tstep_Q = T0, Done = 0, All Regs = 0	Resetn = 0, Clock = running	Asynchronous Reset	System Control	1.1	**processor_top**
Enable Check: Verifies the FSM waits precisely for the Run signal before fetching DIN.	Tstep_Q = T0, Done = 0	Resetn = 1, Run = 0	Idle State Hold		1.2	
Simple immediate load. Proves DIN_out routing through the main Bus MUX.	R0 = 5, Done = 1	Run = 1 (at T0), DIN = 5 (at T1)	mvi R0, 5	Basic Datapath	1.3	
Register to Register copy. Proves shared Bus has no internal data collisions.	R1 = 5, Done = 1	Run = 1 (at T0)	mv R1, R0		1.4	
Standard addition. Proves the shared ALU processes data correctly.	R0 = 10 (at T3), Done=1	Run = 1 (at T0)	add R0, R1	Arithmetic	1.5	
Self-subtraction. Verifies ALU produces absolute zero without logic noise.	R0 = 0 (at T3), Done = 1	Run = 1 (at T0)	sub R0, R0		1.6	
Max Bound: Loads the maximum 9-bit unsigned value to stress the datapath width.	R2 = 511(9'b111111111)	Run = 1 (at T0), DIN = 511 (at T1)	mvi R2, 511	Hardware Limits	1.7	
Adder Tree Stress: R2 is all 1s (from test 1.7). Stretches the parallel adder tree to its physical max count.	R4 = 9 (at T2), Done = 1	Run = 1 (at T0)	ones R2, R4	Ones	1.8	
Adder Tree Normal: Counts ones in 507 (111111011). Verifies the parallel loop logic is accurate.	R3=507 R5 = 8 (at T2) Done = 1	Run = 1 (at T0), DIN = 507 (at T1) (Wait for Done) Run = 1 (at T0)	mvi R3, 507 ones R3, R5		1.9	
Adder Tree Stress: R6 is all 0s. Stretches the parallel adder tree to its physical min count.	R6=0 R7 = 0 (at T2) Done = 1	Run = 1 (at T0), DIN = 0 (at T1) (Wait for Done) Run = 1 (at T0)	mvi R6, 0 ones R6, R7		1.10	
Critical Edge Case: 146 * 3.5. Proves the 11-bit casting prevents intermediate overflow during (146<<2).	R6=146 R7 = 511 (at T2) Done = 1	Run = 1 (at T0), DIN =146 (at T1) (Wait for Done) Run = 1 (at T0)	mvi R6, 146 specialMult R6, R7	Special Mult	1.11	
Critical Safety: Verifies asynchronous reset aborts active execution before completion without data corruption.	Tstep_Q drops to T0 instantly. Done = 0. R3 is NOT updated.	Start add R3, R2. Wait to reach T2. Assert Resetn = 0	Mid-Flight Reset	System Control	1.12	